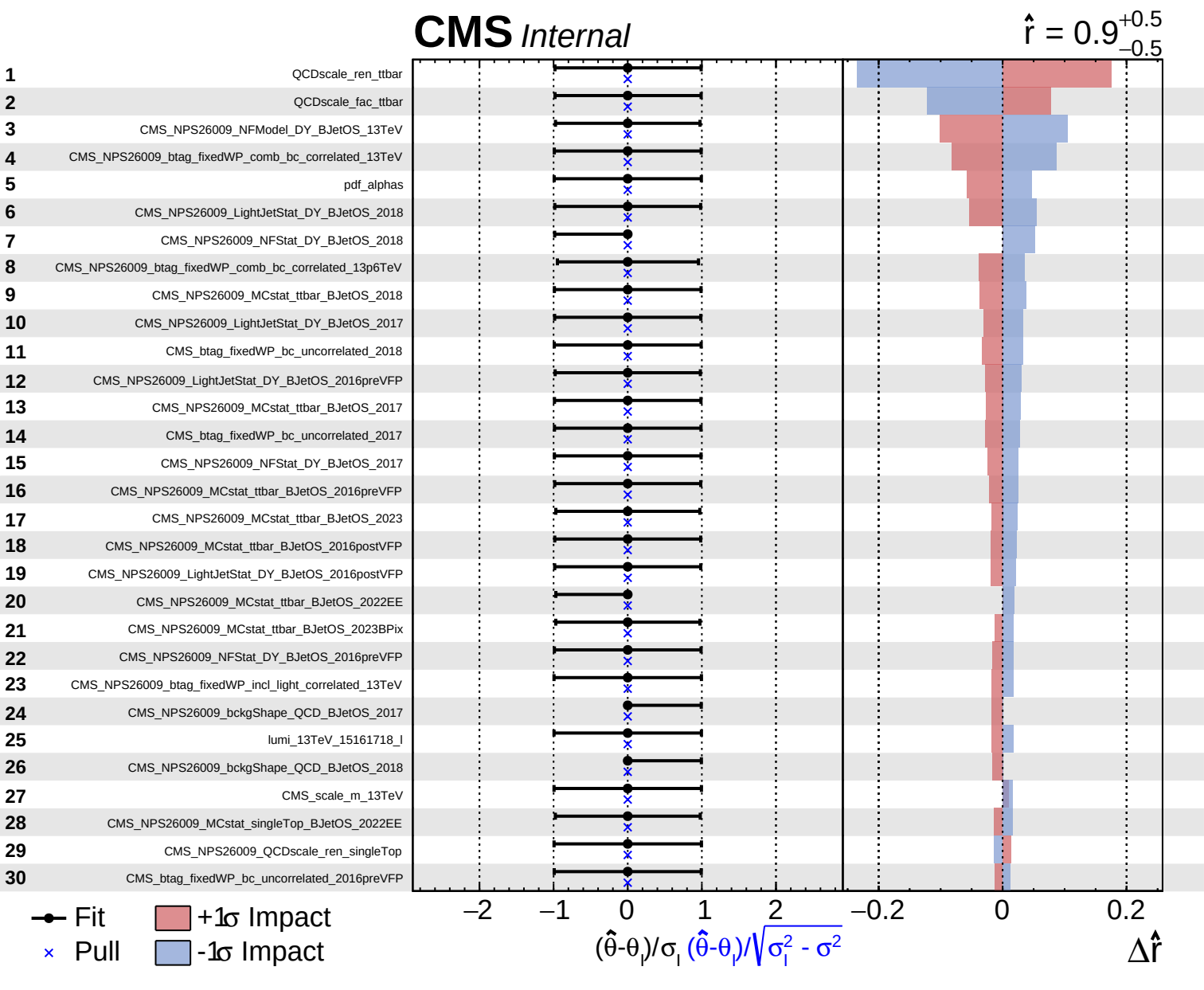
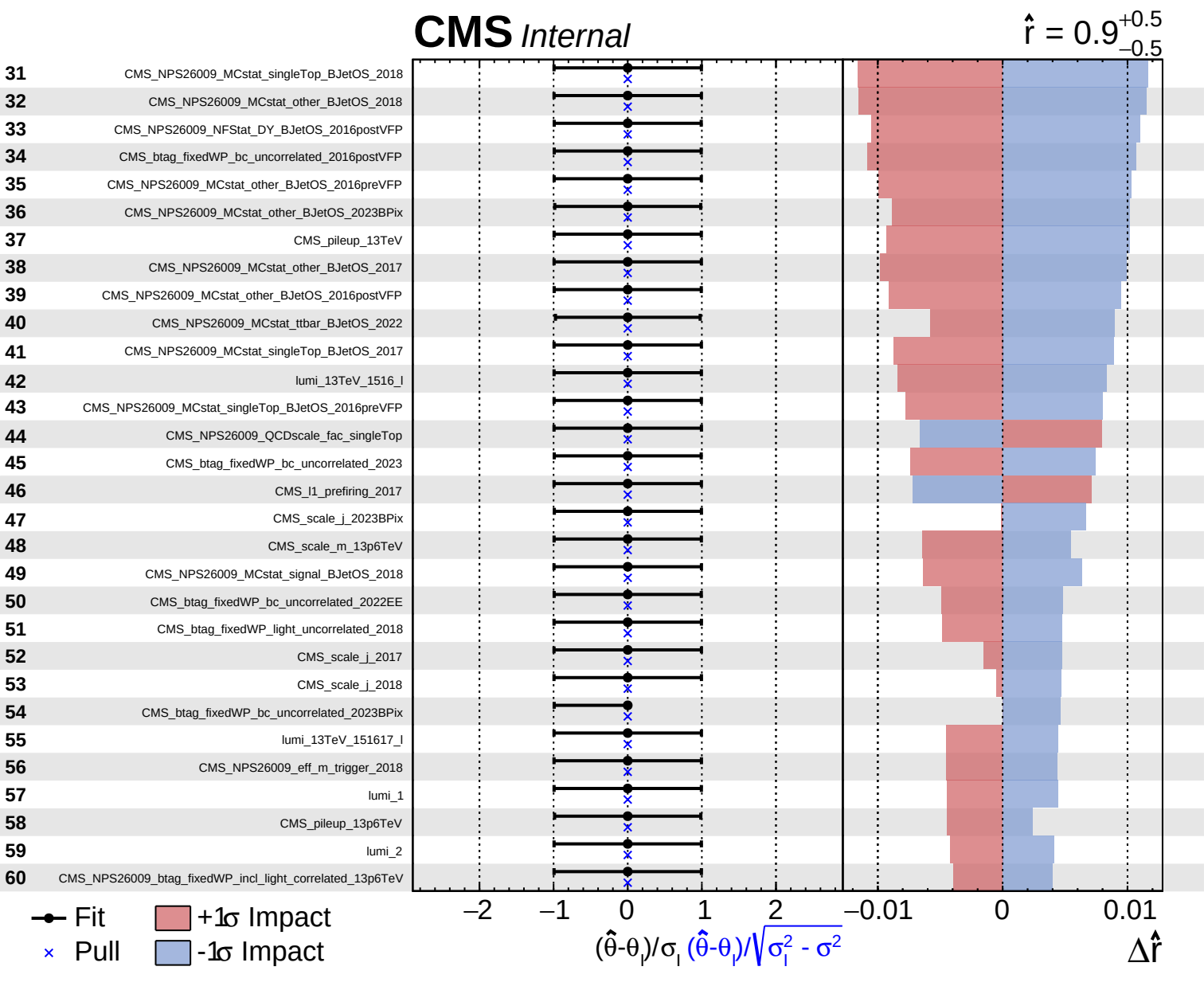
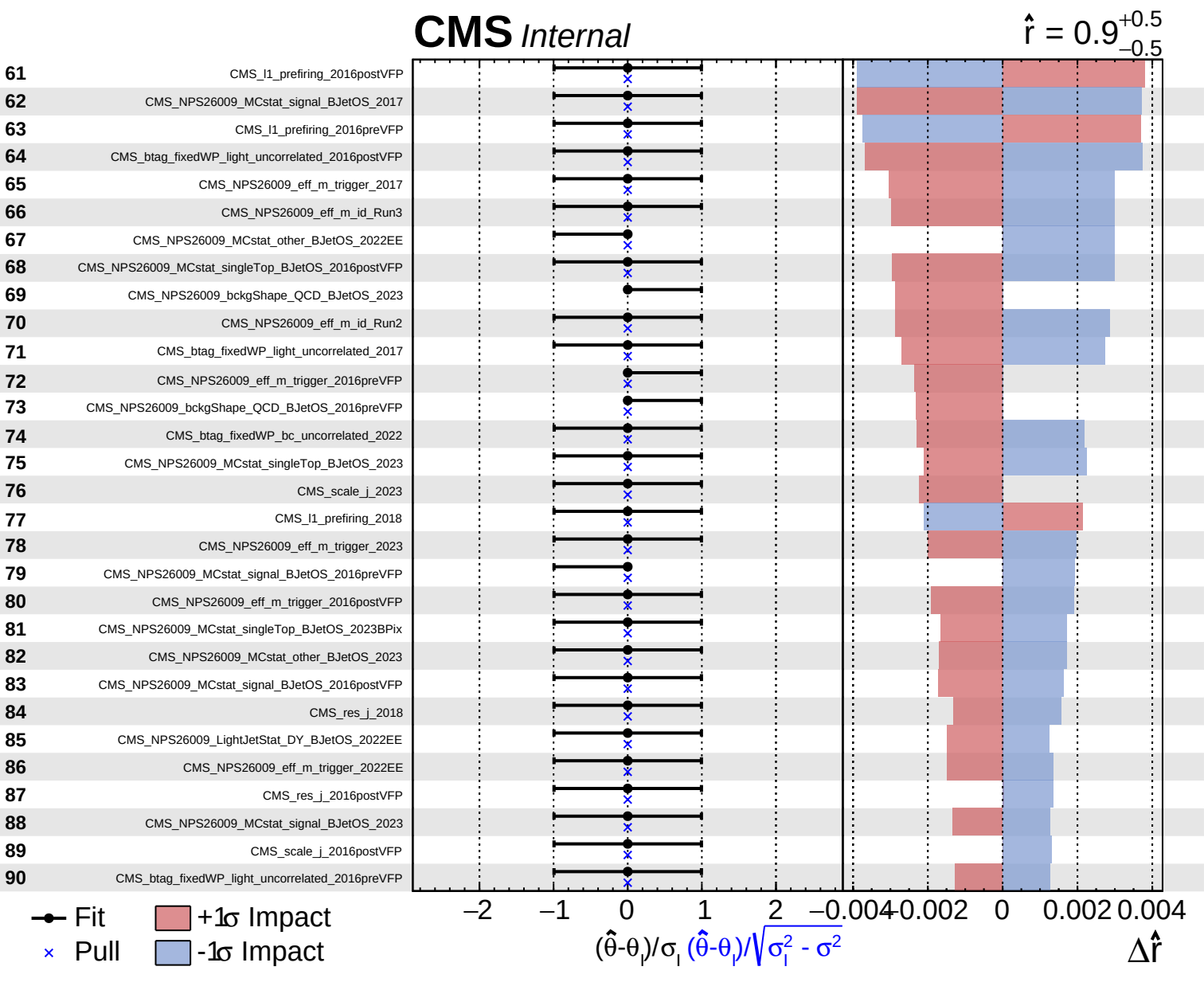
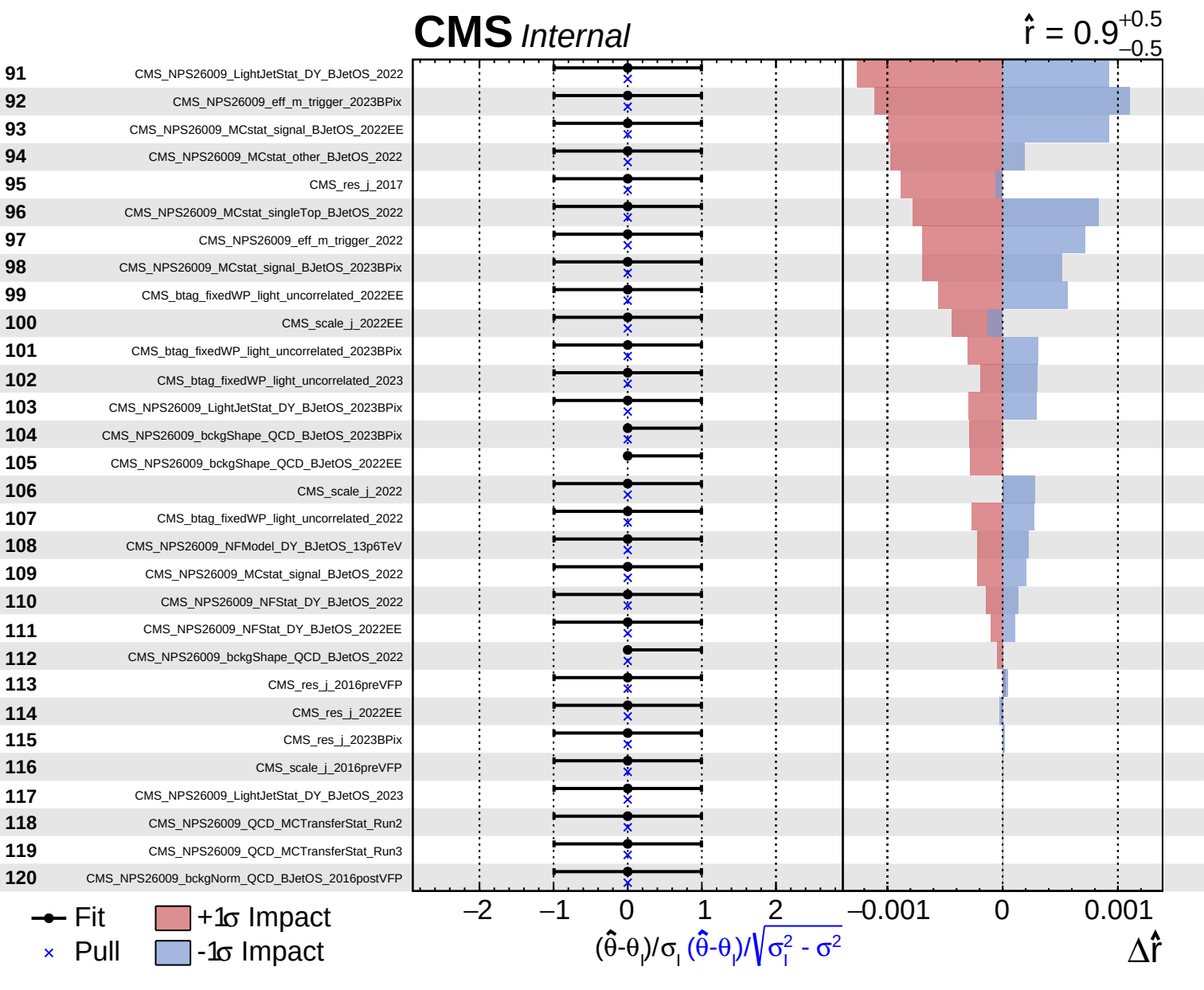
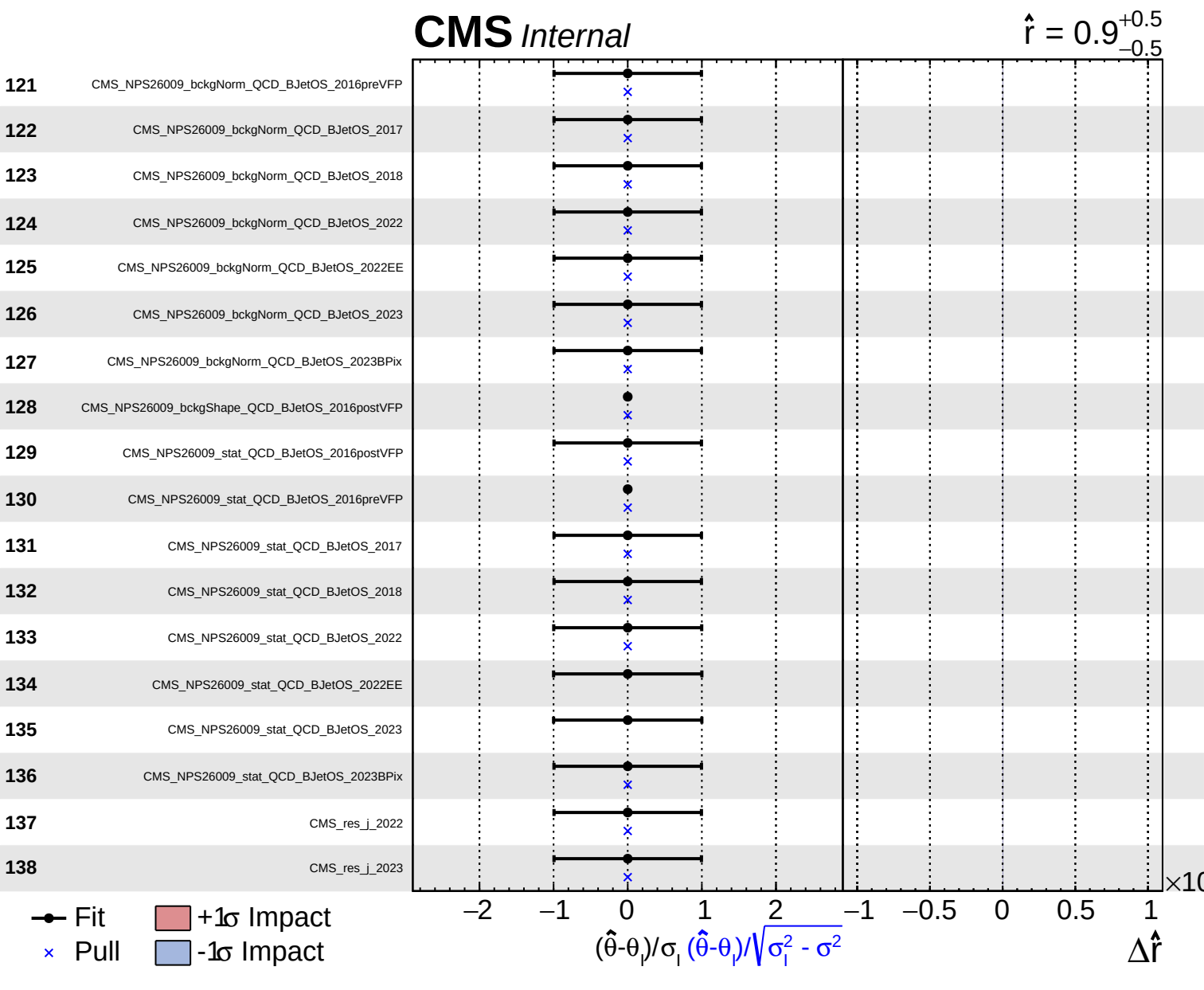

CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
1 QCDscale_ren_ttbar
2 QCDscale_fac_ttbar
3 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
4 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
5 pdf_alphas
6 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
7 CMS_NPS26009_NFStat_DY_BJetOS_2018
8 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
9 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
10 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
11 CMS_btag_fixedWP_bc_uncorrelated_2018
12 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
13 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
14 CMS_btag_fixedWP_bc_uncorrelated_2017
15 CMS_NPS26009_NFStat_DY_BJetOS_2017
16 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
17 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
18 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
19 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
20 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
21 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
22 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
23 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
24 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
25 lumi_13TeV_15161718_l
26 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
27 CMS_scale_m_13TeV
28 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
29 CMS_NPS26009_QCDscale_ren_singleTop
30 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.2 0 0.2
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
31 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
32 CMS_NPS26009_MCstat_other_BJetOS_2018
33 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
34 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
35 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
36 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
37 CMS_pileup_13TeV
38 CMS_NPS26009_MCstat_other_BJetOS_2017
39 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
40 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
41 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
42 lumi_13TeV_1516_l
43 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
44 CMS_NPS26009_QCDscale_fac_singleTop
45 CMS_btag_fixedWP_bc_uncorrelated_2023
46 CMS_l1_prefiring_2017
47 CMS_scale_j_2023BPix
48 CMS_scale_m_13p6TeV
49 CMS_NPS26009_MCstat_signal_BJetOS_2018
50 CMS_btag_fixedWP_bc_uncorrelated_2022EE
51 CMS_btag_fixedWP_light_uncorrelated_2018
52 CMS_scale_j_2017
53 CMS_scale_j_2018
54 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
55 lumi_13TeV_151617_l
56 CMS_NPS26009_eff_m_trigger_2018
57 lumi_1
58 CMS_pileup_13p6TeV
59 lumi_2
60 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_l)/\sigma_l$
−0.01
0
0.01
$(\hat{\theta}-\theta_l)/\sqrt{\sigma_l^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
61 CMS_l1_prefiring_2016postVFP
62 CMS_NPS26009_MCstat_signal_BJetOS_2017
63 CMS_l1_prefiring_2016preVFP
64 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
65 CMS_NPS26009_eff_m_trigger_2017
66 CMS_NPS26009_eff_m_id_Run3
67 CMS_NPS26009_MCstat_other_BJetOS_2022EE
68 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
69 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
70 CMS_NPS26009_eff_m_id_Run2
71 CMS_btag_fixedWP_light_uncorrelated_2017
72 CMS_NPS26009_eff_m_trigger_2016preVFP
73 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
74 CMS_btag_fixedWP_bc_uncorrelated_2022
75 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
76 CMS_scale_j_2023
77 CMS_l1_prefiring_2018
78 CMS_NPS26009_eff_m_trigger_2023
79 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
80 CMS_NPS26009_eff_m_trigger_2016postVFP
81 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
82 CMS_NPS26009_MCstat_other_BJetOS_2023
83 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
84 CMS_res_j_2018
85 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
86 CMS_NPS26009_eff_m_trigger_2022EE
87 CMS_res_j_2016postVFP
88 CMS_NPS26009_MCstat_signal_BJetOS_2023
89 CMS_scale_j_2016postVFP
90 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.004
−0.002
0
0.002
0.004
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
91 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
92 CMS_NPS26009_eff_m_trigger_2023BPix
93 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
94 CMS_NPS26009_MCstat_other_BJetOS_2022
95 CMS_res_j_2017
96 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
97 CMS_NPS26009_eff_m_trigger_2022
98 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
99 CMS_btag_fixedWP_light_uncorrelated_2022EE
100 CMS_scale_j_2022EE
101 CMS_btag_fixedWP_light_uncorrelated_2023BPix
102 CMS_btag_fixedWP_light_uncorrelated_2023
103 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
104 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
105 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
106 CMS_scale_j_2022
107 CMS_btag_fixedWP_light_uncorrelated_2022
108 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
109 CMS_NPS26009_MCstat_signal_BJetOS_2022
110 CMS_NPS26009_NFStat_DY_BJetOS_2022
111 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
112 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
113 CMS_res_j_2016preVFP
114 CMS_res_j_2022EE
115 CMS_res_j_2023BPix
116 CMS_scale_j_2016preVFP
117 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
118 CMS_NPS26009_QCD_MCTransferStat_Run2
119 CMS_NPS26009_QCD_MCTransferStat_Run3
120 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.001
0
0.001
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
121 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
122 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
123 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
124 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
125 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
126 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
127 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
128 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
129 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
130 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
131 CMS_NPS26009_stat_QCD_BJetOS_2017
132 CMS_NPS26009_stat_QCD_BJetOS_2018
133 CMS_NPS26009_stat_QCD_BJetOS_2022
134 CMS_NPS26009_stat_QCD_BJetOS_2022EE
135 CMS_NPS26009_stat_QCD_BJetOS_2023
136 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
137 CMS_res_j_2022
138 CMS_res_j_2023
×10
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$